Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

Manual Volume II — Matter I: Elements 1–118

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October 2026

$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.

Nuff said.

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1 Manual Volume II Scope and Record Discipline

This manual volume carries the **Matter I: Elements 1–118** sector execution records, coord-0 support-enclosure tables, fusion-closure and exoshedding ledgers, and audit fixtures downstream from the main AΩD note [4]. The main note carries the AF/AFC/AΩD ontology and calculus. This manual volume carries elemental calculation records, concentric enclosure setup diagrams, D.E.C. row invariants, quarantined external-comparison templates, and audit tables.

1.1 Reading route and calculation spine

Follow Main in its dependency order: specific distinction (Main §1), closure and recurring wave (Main §2), curl/duon/field folding (Main §3), complete wave state and relational time (Main §4), and finite Stokes/SADAR/enclosure/fusion values (Main §5). Every record in this volume declares

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega),$$

forms retained cycle provenance $C_f[\Sigma]_{A\leftrightarrow\Omega}$, evaluates $\mathsf{AO}_\mu(C) = \langle \Pi_\alpha, H_\mu, \Pi_\Omega \rangle$, and computes $\mathfrak{S}_B(C) = p^D_{C,A\leftrightarrow\Omega} A_{C,A\leftrightarrow\Omega}$ and boundary accumulation $\Phi^D_{\partial_T R} = \sum_{e\in\partial_T R} \mathfrak{S}_B(C_e)$. Tables store evaluated values; tables are not the calculus.

1.2 Exact arithmetic and external-comparison quarantine

All AΩ-internal quantities in this volume—support cardinalities $\Pi(p{:}q)$, reflection durations RD, group boundaries Γ_m, fusion durations RD_{fuse}, pressures P^D, closure compatibilities C_{close}, residuals S_{n+1}, surpluses X_{shedding}, and occupancy fractions—are evaluated in exact integers ($\mathbb{Z}$) and exact rationals ($\mathbb{Q}$). External constants, measured-sector units, and measured atomic/spectral targets remain strictly quarantined inside $\mathsf{Q}_{\text{ext}}(\nu_{Z,A\leftrightarrow\Omega}; \mathcal{T}_Z)$ (App. B).

Specification discipline. A Matter I record may use a compact curling-curl specification $a{:}b{:}c :: L$ only after its construction trace $HT(K; B)$ is declared or inherited (Main §3.5, (37)).

2 Scope and Imported Mathematical Spine

2.1 Downstream sector scope and calculation spine

Setup. This note instantiates the foundational calculus of Main v41.02 [4] on the coord-0 elemental support ladder $Z = 1 \ldots 118$ (together with the unretained open cutoff at $Z = 119 \ldots 126$ on C_{22}). No new axioms or primitive relations are introduced. Every calculation begins at the first finite anchor $a_0 = \mathtt{00}_8$ (Main (3)) with local origin $0 \vdash$, declares a tesseract-admissible boundary and integer comparison window

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega), \tag{1}$$

and executes the active $A\leftrightarrow\Omega$ calculation spine (Main §5.13, Fig. 11; Manual §1.3):

$$C_f[\Sigma]_{A\leftrightarrow\Omega}, \qquad \mathsf{AO}_\mu(C) = \langle \Pi_\alpha(C), H_\mu(C), \Pi_\Omega(C) \rangle, \qquad \mathfrak{S}_B(C) = p^D_{C,A\leftrightarrow\Omega}(B) A_{C,A\leftrightarrow\Omega}(B), \tag{2}$$

with boundary accumulation

$$\Phi^D_{\partial_T R} = \sum_{e\in\partial_T R} \mathfrak{S}_B(C_e). \tag{3}$$

Tables in this note store evaluated integer and rational values produced by (1)–(3); tables are document containers, not the calculus.

Derivation. By Main (1)–(5), Null sameness ($\mathcal{N} \equiv \infty,\ \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$) is schematic idempotency; once a finite distinction commits an event on $Q_4 = \{0,1\}^4$ (Main (8)), its identity, lineage, and branch/hinge/counterbranch order $M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon)$ remain in the trace. By Main (12)–(16), a certified closure $\mathrm{Cert}(C)$ requires at least two committed phase-bearing traversals ($m(C) \geq 2$) together with its retained-start return witness $\mathcal{W}_C$, contributing one closure-phase cycle $\Delta\Phi(C) = 1_C$.

Specialization. In the $D0$ exact internal fixture layer of this note, all support cardinalities Π, reflection durations RD, channel pressures P^D, closure compatibilities C_{close}, rung residuals S_{n+1}, and sheddic surpluses X_{shedding} are exact integers in $\mathbb{Z}$. Measured-sector element symbols (H … Og for $Z = 1 \ldots 118$) are application-layer carrier labels attached only after the coord-0 construction is admitted (Main §5.17.10).

Calibration. Each elemental rung binds one integer coord-0 index $Z \in \{1, \ldots, 118\}$ and one reference nucleon count $A \in \mathbb{Z}_{\geq 1}$ in the count basis $\mathcal{B}_{\mathrm{curl}} = (3{:}3, 3{:}4, 1{:}2)$ (Main App. B.4.1), with collision/channel capacity $C_3 = 3$ (Main (77), Manual §5.4). External mass or spectral target comparisons remain quarantined in $\mathsf{Q}_{\mathrm{ext}}(\nu_{x,A\leftrightarrow\Omega}; \mathcal{T}_x)$ (Manual §1.8, App. B).

Falsification. Importing an undeclared external constant into an $A\leftrightarrow\Omega$-internal calculation, collapsing Null $\mathcal{N}$, fibre zero 0, vertex 0000, anchor $\mathtt{00}_8$, and scope B_0 into a single type, or using a table label without a bound closure trace $HT(K;B)$ contradicts (1)–(3).

Provenance. Main (1)–(5), Main (8), Main (11)–(16), Main (40), Main (44), Main (92).

Literature note. Foundational protocol and calculus: Poyau [2, 3, 4].

2.2 Curling-curl support and branch/hinge/branch reflection duration

Setup. A support expression $\Sigma = (s_1, \ldots, s_m; L)$ (Main Def. 8, (36)) with clauses $s_i = p_i{:}q_i$ and optional shared enclosure qualifier L proposes a curling-curl support specification K_{prop}. Admission on $B = (\partial_T F, \omega)$ is the traced calculation (Main Def. 9, (37))

$$K_{\mathrm{prop}} \xrightarrow{HT(K;B)} K_{\mathrm{adm}}(B), \tag{4}$$

binding the complete motif record $K_{\mathrm{adm}}(B, i; \mathrm{cert})$ of Main (33). For each curling-curl clause $p{:}q$, import the finite edge-shell walk support model of Main §3.5, (35) and Main App. H.5, (168):

$$\Pi(p{:}q) = q^p + q, \qquad p, q \in \mathbb{Z}_{\geq 1}. \tag{5}$$

Here q^p counts memoryless q-direction walk branches and $+q$ is the retained Hamming-1 edge-shell term on Q_4.

Derivation. For a certified cycle C on B, the Alpha–Omega hinge kernel (Main Def. 14, (44)) has branch/hinge/branch reflection duration (Main §3.3.2, (31), Main §3.9, (49), and Main App. H.5, (169)):

$$\mathsf{AO}_\mu(C) = \langle \Pi_\alpha(C), H_\mu(C), \Pi_\Omega(C) \rangle, \qquad RD_{A\leftrightarrow\Omega}(C; B) = |\Pi_\alpha(C)| + H_\mu(C) + |\Pi_\Omega(C)|. \tag{6}$$

Under symmetric branch supports ($|\Pi_\alpha(C)| = |\Pi_\Omega(C)| = \Pi(p{:}q)$), (6) gives $RD_{A\leftrightarrow\Omega}(p{:}q; B) = 2\Pi(p{:}q) + H_\mu(B)$ (Main (32)). Under the single-hinge reduction $H_\mu(B) = 1$ (Main (50), (170)),

$$RD^{(1)}_{A\leftrightarrow\Omega}(p{:}q) = 2\Pi(p{:}q) + 1 = 2(q^p + q) + 1. \tag{7}$$

Specialization. Evaluating (5) and (7) on the canonical curling-curl families of Main App. B.2, Table 1 yields the exact integer primitives in Table 1. Note the strict typing separation of Main (34) and Main App. B.1.10: Dimonon = 2 Monon ≠ Dion, and tetrion (bare four-cycle), tetron (E_4 four-seat inner enclosure), tetratrion (4 × trion = 3:4), and tetrad (seated tetron closure) are distinct types.

Table 1: **Imported curling-curl support primitives and single-hinge reflection durations (Main §3.5, Main App. B.2, Main App. H.5).**

Form $p{:}q$	Family name	Stem	Meaning	$\Pi(p{:}q) = q^p + q$	$RD^{(1)}_{A\leftrightarrow\Omega} = 2\Pi + 1$	Default hexon instance (C_6)
1:2	Dimonon	Dimon	2 × monon	$2^1 + 2 = 4$	$2(4) + 1 = 9$	1:2:6 = Dimonhexon
3:1	Trio	Trio	1 × trion	$1^3 + 1 = 2$	$2(2) + 1 = 5$	3:1:6 = Triohexon
3:2	Ditrion	Ditrio	2 × trion	$2^3 + 2 = 10$	$2(10) + 1 = 21$	3:2:6 = Ditriohexon
3:3	Tritrion	Tritrio	3 × trion	$3^3 + 3 = 30$	$2(30) + 1 = 61$	3:3:6 = Tritriohexon
3:4	Tetratrion	Tetratrio	4 × trion	$4^3 + 4 = 68$	$2(68) + 1 = 137$	3:4:6 = Tetratriohexon

Calibration. The count compression basis of Main App. B.4.1 fixes the ordered triple

$$\mathcal{B}_{\text{curl}} = (3{:}3,\ 3{:}4,\ 1{:}2) = (\text{Tritrio},\ \text{Tetratrio},\ \text{Dimon}), \tag{8}$$

so that any count form $a{:}b{:}c :: L$ expands exactly as

$$a{:}b{:}c :: L \ \equiv\ \text{Tritrio}^a :: \text{Tetratrio}^b :: \text{Dimon}^c :: \text{out}(L). \tag{9}$$

In particular, the basic shared hexon is $1{:}1{:}0 :: 6$ = Tritrio :: Tetratrio :: hexon ≡ $(3{:}3)^1$ :: $(3{:}4)^1$ :: $(1{:}2)^0$ :: 6 (Main App. B.4.2).

Falsification. Invoking $RD_{A\leftrightarrow\Omega}(C_f[p{:}q])$ before declaring the certified cycle compression $C_f[p{:}q]_{A\leftrightarrow\Omega}$ violates the cycle audit (Main §5.17.7). Treating $RD = 2\Pi + 1$ as unconditional when $H_\mu \neq 1$ or when branches are asymmetric violates Main (31)–(32).

Provenance. Main (31)–(37), Main (40), Main (44)–(50), Main (168)–(171), Main App. B.1–B.4.

Literature note. Combinatorial walk support and tesseract encoding: Poyau [4] (Main §1.4, Main App. H).

3 Seat Compatibility, Enclosures, and Coord-0 Grouping

3.1 Even enclosure sequence and cumulative coord-0 boundaries

Setup. Import the seat-compatibility and shell/enclosure routing hierarchy of Main §5.7.1–§5.7.13 and Main §5.8:

1. **Current-side double forms (Main §5.7.2–§5.7.3):** E_2 = duon, E_2^0 = open-seat duon, and $E_2^\pm$ = duad (the seat-taken state of E_2).

2. **Default inner enclosure (Main §5.7.4):** E_4 = tetron, the default four-seat inner enclosure anchored at local origin $0\vdash$.
3. **Outer even support enclosures (Main §5.7.5–§5.7.9, Main App. B.3):** C_{2m} denotes the outer support enclosure of even order $2m$ for $m \geq 3$, beginning with C_6 = hexon ($\mathsf{H}_3 + \mathsf{H}_3 = 6$), C_8 = octon, C_{10} = decon, and continuing through higher even orders C_{2m} ($m > 5$).

Derivation. At coordinate 0, a coord-0 shell-stack declaration has the canonical form (Main §5.7.13)

$$0\vdash:: Z : A : 0 : L\langle a : b : c :: L\rangle, \tag{10}$$

and a filled shared support enclosure of order $2m$ ($m \geq 3$) has the canonical form (Main §5.7.10, §5.7.13, App. B.4.3)

$$0\vdash:: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle \;=\; \text{Tritrio}^m :: \text{Tetratrio}^m :: \text{out}(2m). \tag{11}$$

Summing the even seat capacities $2k$ from the first outer enclosure $k = 3$ (C_6) through $k = m$ yields the cumulative coord-0 group boundary (Main §5.7.13):

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0. \tag{12}$$

The strict interval condition

$$\Gamma_{m-1} < Z \leq \Gamma_m \tag{13}$$

uniquely selects the coord-0 support-enclosure group C_{2m} and local seat index $s = Z - \Gamma_{m-1} \in \{1, \ldots, 2m\}$ for every integer $Z \geq 1$. Along each ring C_{2m}, the $2m$ seats partition into:

$$a(s) = \left\lceil \frac{s}{2} \right\rceil, \qquad b(s) = \left\lfloor \frac{s}{2} \right\rfloor, \qquad c(s) = s \bmod 2, \tag{14}$$

so that $s = 2$ on C_6 gives 1:1:0 :: 6 (Main §5.8.1), odd seats $s = 2j - 1$ carry one open-seat duon ($c = 1, E_2^0$), even seats $s = 2j$ close j seated duads ($a = b = j, c = 0, E_2^{\pm}$), half-shell turnaround occurs at $s = m$ on the 6 o'clock H_μ hinge axis ($\Pi_\alpha \to \Pi_\Omega$), and $s = 2m$ ($Z = \Gamma_m$) locks the filled shared enclosure $m{:}m{:}0 :: 2m$ on the 12 o'clock spine.

Specialization. Evaluating (12)–(14) across $m = 3 \ldots 11$ yields the nine concentric outer support enclosures $\mathsf{C}_6 \ldots \mathsf{C}_{22}$ summarized in Table 2 and rendered in Fig. 1. Across $m = 3 \ldots 11$, total coord-0 capacity is $\Gamma_{11} = 11(12) - 6 = 126$ slots; the admitted elemental ladder $Z = 1 \ldots 118$ occupies the exact rational fraction $118/126 = 59/63$ of Γ_{11} (and $14/22 = 7/11$ on C_{22}), while slots $Z = 119 \ldots 126$ on C_{22} form the unretained open cutoff ($\text{TTL}_{\mathcal{C}}(\partial F, \omega) < |\omega|$).

Table 2: **Coord-0 support-enclosure groups across $m = 3 \ldots 11$ (Main §5.7.10–§5.7.13, Main §5.8).**

m	Encl.	Name	$2m$	Γ_{m-1}	Γ_m	Z range	H_μ turn ($s = m$)	Anchor ($Z = \Gamma_m$)	Filled $m{:}m{:}0 :: 2m$	C_{close}
3	C_6	hexon	6	0	6	1 … 6	$Z = 3$ (Li)	$Z = 6$ (C)	3:3:0 :: 6	18
4	C_8	octon	8	6	14	7 … 14	$Z = 10$ (Ne)	$Z = 14$ (Si)	4:4:0 :: 8	24
5	C_{10}	decon	10	14	24	15 … 24	$Z = 19$ (K)	$Z = 24$ (Cr)	5:5:0 :: 10	30
6	C_{12}	C_{12}	12	24	36	25 … 36	$Z = 30$ (Zn)	$Z = 36$ (Kr)	6:6:0 :: 12	36
7	C_{14}	C_{14}	14	36	50	37 … 50	$Z = 43$ (Tc↑)	$Z = 50$ (Sn)	7:7:0 :: 14	42
8	C_{16}	C_{16}	16	50	66	51 … 66	$Z = 58$ (Ce)	$Z = 66$ (Dy)	8:8:0 :: 16	48
9	C_{18}	C_{18}	18	66	84	67 … 84	$Z = 75$ (Re)	$Z = 84$ (Po↑)	9:9:0 :: 18	54
10	C_{20}	C_{20}	20	84	104	85 … 104	$Z = 94$ (Pu↑)	$Z = 104$ (Rf↑)	10:10:0 :: 20	60
11	C_{22}	C_{22}	22	104	126	105 … 126*	$Z = 115$ (Mc↑)	$Z = 126$ (−−)	11:11:0 :: 22	66

*On C_{22}, $Z = 105 \ldots 118$ (Db … Og) are admitted exoshedding/yro routes; $Z = 119 \ldots 126$ are unretained open cutoff slots.

Calibration. By Main §5.2.4, (90) and Main §5.17.4, field-count scaling $\mathcal{O}_{\partial F}(\mathcal{B}) = n_F(\mathcal{B}) \log n_F(\mathcal{B})$ (for $n_F(\mathcal{B}) \geq 1$) and seat-compatible shell/enclosure routing ((10)–(14)) are distinct calculations on the same declared boundary/window:

$$\begin{aligned} B \to \mathcal{F}_B \to n_F(B) \to \mathrm{ADAR}_{i \leftarrow j} \to \mathcal{O}_{\partial F}(B) \to \mathrm{SADAR}, \\ \mathsf{C}_{2m} \to X_{\mathrm{shedding}} \to \mathrm{SheddicPath}(X_{\mathrm{shedding}}). \end{aligned} \tag{15}$$

Falsification. Assigning an element Z to an odd support enclosure, shifting Γ_m away from $m(m+1)-6$, or conflating $n_F \log n_F$ field-count weighting with C_{2m} seat routing fails the seat/enclosure routing audit (Main §5.17.4).

Provenance. Main §5.2.4 (89)–(90), Main §5.7.1–§5.7.13 (112)–(113), Main §5.8.1–§5.8.2, Main App. B.3–B.4.

Literature note. Coord-0 support-enclosure ladder and nested shell-stack pattern: Poyau [4] (Main §5.7–§5.8).

4 Fusion Closure, Rung Residuals, and Complete Contact Ledger

4.1 Fusion closure, rung residuals, and exoshedding lift

Setup. Let F and G be admitted curling-curl support forms on $B = (\partial_T F, \omega)$ with certified cycle-valued compressions $\widehat{F} = \mathrm{Compress}(F)$ and $\widehat{G} = \mathrm{Compress}(G)$ (Main Def. 13, (40)). Import the fusion closure operator of Main §5.9.1, (114):

$$\mathrm{Fuse}_B(F, G) := \mathrm{Close}_B \left(\widehat{F} \, \mu \, \widehat{G} \right). \tag{16}$$

Closed or reclosed admitted forms fuse; exposed fragments or open proposals without bound inner certificates are outside the domain of (16).

Derivation. By Main (115), the branch/hinge/branch fusion duration across the shared hinge μ is

$$RD_{\mathrm{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu, \widehat{F}, \widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{17}$$

For each elemental closure rung $n+1$ ($Z = 1 \ldots 118$, plus cutoff slots $119 \ldots 126$ on C_{22}), the rung closure, signed rung residual, and sheddic surplus are (Main §5.9.1, (116)–(118)):

$$C^{\mathrm{ret}}_{n+1} = \mathrm{Close}_{B_{n+1}} \left(\widehat{F}_n \, \mu \, \widehat{G}_n \right), \qquad S_{n+1} = P^{D}_{n+1} - C_{\mathrm{close},n+1}, \qquad X_{\mathrm{shedding},n+1} = \max(0, S_{n+1}). \tag{18}$$

By Main §5.6.4, (107)–(108) and Main §5.9.1, the sign of S_{n+1} partitions the rungs into three distinct operational regimes:

1. **Neutral / Seated Closure ($S_{n+1} = 0 \implies X_{\mathrm{shedding},n+1} = 0$):** Local duonic pressure balances enclosure closure compatibility $C_{\mathrm{close},n+1} = 3(2m)$, and seat-compatible retention satisfies $\mathrm{TTL}_{\mathcal{C}}(\partial F, \omega) \geq |\omega|$ (Main (113)).
2. **Deficit / Open-Seat Acceptor ($S_{n+1} < 0 \implies X_{\mathrm{shedding},n+1} = 0$):** A negative closure residual is a deficit/open check (E_2^0 open-seat acceptor), never sheddic surplus (Main §5.6.4).

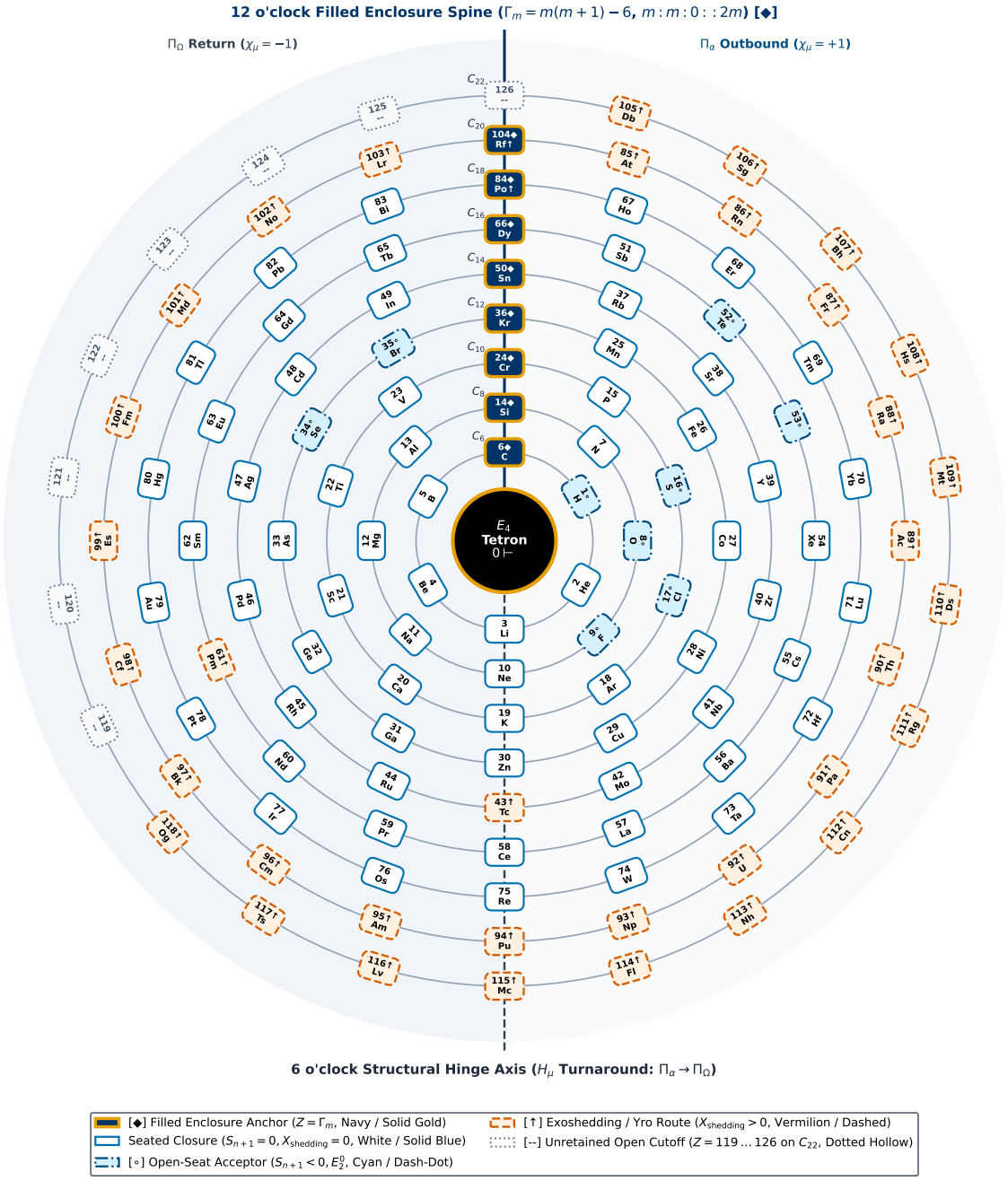


Figure 1: **Concentric coord-0 support-enclosure chart ($Z = 1 \ldots 118$).** The default four-seat inner enclosure E_4 (tetron) anchors the concentric outer even support enclosures C_{2m} ($m = 3 \ldots 11$, orders $C_6 \ldots C_{22}$) partitioned by $\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6$ with $\Gamma_{m-1} < Z \leq \Gamma_m$. Filled shared enclosures $0 \vdash :: Z : A : 0 : 2m \langle m : m : 0 :: 2m \rangle$ lock along the 12 o'clock spine (⧫), while half-shell branch turnarounds ($\Pi_\alpha \to \Pi_\Omega$) align along the 6 o'clock H_μ hinge axis. Seated closures ($X_{\text{shedding}} = 0$, solid blue), open-seat acceptors (E_2^0, dash-dot cyan ∘), exoshedding/yro routes ($X_{\text{shedding}} > 0$, dashed vermilion ↑), and the unretained open cutoff at $Z = 119 \ldots 126$ on C_{22} (dotted hollow) use redundant glyph and stroke encoding.

3. **Surplus / Exoshedding Route ($S_{n+1} > 0 \implies X_{\text{shedding},n+1} = S_{n+1} > 0$):** Positive surplus is placed on a declared sheddic path (Main §5.6.5, (109)–(110))

$$\text{SheddicPath}(X_{\text{shedding}}) \in \{E_2, \mathsf{C}_6, \mathsf{C}_8, \mathsf{C}_{10}, \ldots, \mathsf{C}_{2n}\}, \qquad X_{\text{shedding}} = X_{\text{endo}} + X_{\text{exo}} + X_{\text{redir}} + X_{\text{open}}. \tag{19}$$

When $X_{\text{exo},r} > 0$ at scope B_r, exoshedding routes surplus support to the next enclosure scope via the lift and higher-scope fusion chain (Main §5.9.2, (119)–(121)):

$$K^{r+1}_{\text{prop}} = \text{Lift}_{B_r \to B_{r+1}}(X_{\text{exo},r}) \xrightarrow{HT(K;B_{r+1})} K^{r+1}_{\text{adm}}(B_{r+1}), \qquad C^{\text{ret}}_{r+1} = \text{Close}_{B_{r+1}}\left(\widehat{F}_r \; \mu \; K^{r+1}_{\text{adm}}(B_{r+1})\right). \tag{20}$$

When instability occurs across an odd window length $L = 2k + 1$ between adjacent even support enclosures C_{2k} and C_{2k+2}, it is recorded by the odd-window route-status suffix $\text{Yro}_L(K; B)$ (Main §5.6.6).

Specialization. For each rung Z with reference nucleon count A on enclosure C_{2m} (seat $s = Z - \Gamma_{m-1}$):

$$\Pi^{-}_{\widehat{F}}(B_Z) = 30Z + 68(A - Z), \qquad H_{\mu,\widehat{F},\widehat{G}}(B_Z) = 1, \qquad \Pi^{+}_{\widehat{G}}(B_Z) = 4s, \tag{21}$$

where $\Pi(3{:}3) = 30$ (Tritrio), $\Pi(3{:}4) = 68$ (Tetratrio), and $\Pi(1{:}2) = 4$ (Dimon). Substituting (21) into (17) yields the exact integer fusion duration $RD_{\text{fuse}}(Z, A, s) = 30Z + 68(A - Z) + 1 + 4s$.

Calibration. Across $Z = 1 \ldots 118$, the 118 admitted elements partition into:

- **8 Filled Shared Enclosure Anchors ($Z = \Gamma_m \leq 104$):** $Z \in \{6, 14, 24, 36, 50, 66, 84, 104\}$ (C, Si, Cr, Kr, Sn, Dy, Po, Rf), of which $Z \in \{6, 14, 24, 36, 50, 66\}$ are neutral seated anchors ($S_Z = 0$) and $Z \in \{84, 104\}$ carry active exoshedding ($X_{\text{shedding}} > 0$).
- **66 Non-Anchor Seated Retained Closures ($S_Z = 0, X_{\text{shedding}} = 0$):** Retained $E_2^{\pm}$ duad closures with $\text{TTL}_{\mathcal{C}} \geq |\omega|$.
- **9 Open-Seat Acceptors ($S_Z = -3 < 0, X_{\text{shedding}} = 0$):** $Z \in \{1, 8, 9, 16, 17, 34, 35, 52, 53\}$ (H, O, F, S, Cl, Se, Br, Te, I), carrying an unseated E_2^0 deficit channel.
- **37 Exoshedding / Yro Route Elements ($S_Z > 0, X_{\text{shedding}} > 0$):** The two mid-shell hinge-instability elements $Z \in \{43\ (\text{Tc}), 61\ (\text{Pm})\}$ together with the post-bismuth shedding ladder $Z \in \{84\ (\text{Po}) \ldots 118\ (\text{Og})\}$ (including anchors $Z = 84, 104$), each routing positive surplus X_{shedding} into C_{2m+2} with $\text{TTL}_{\mathcal{C}} < |\omega|$.
- **8 Unretained Open Cutoff Slots ($Z = 119 \ldots 126$ on C_{22}):** Slots $s = 15 \ldots 22$ on C_{22} have $\text{TTL}_{\mathcal{C}} = 0 < |\omega|$ and route entirely to unretained open loss X_{open}.

Falsification. Reporting $X_{\text{shedding}} > 0$ for a negative closure residual $S_{n+1} < 0$, omitting a declared SheddicPath when $X_{\text{shedding}} > 0$, or treating an odd-window Yro_L route-status suffix as an even support enclosure name contradicts Main §5.6.4–§5.6.6 and Main §5.17.5.

Provenance. Main §5.6.4–§5.6.6 (107)–(110), Main §5.7.11–§5.7.12 (112)–(113), Main §5.9.1–§5.9.3 (114)–(123).

Literature note. Reference nuclear mass numbers and ground-state shell designations: Wang et al. [5]; NIST ASD [1].

4.2 Complete coord-0 elemental ledger ($Z = 1 \ldots 118$ and C_{22} cutoff $119 \ldots 126$)

Source artifacts. The exact integer ledger rows and audit assertions in Table 3 are generated and verified deterministically by `manual2/scripts/verify_matter1_ledger.py`,

with outputs stored in `manual2/data/m1/matter1_elements_1_118_ledger.csv` and `manual2/data/m1/matter1_audit_report.json`.

Table 3: **Complete coord-0 fusion closure and routing ledger ($Z = 1 \ldots 118$ plus C_{22} cutoff $119 \ldots 126$).** Evaluated in exact integer arithmetic from Main §3.5, Main §5.7.13, and Main §5.9.1.

Z	Sym	A	C_{2m}	$a{:}b{:}c :: L$	Π^-	H_μ	Π^+	RD_{fuse}	P^D	C_{close}	S_{n+1}, X_{shed}	Route / Status
1	H	1	C_6	1:0:1 :: 6	30	1	4	35	15	18	−3, 0	Acceptor E_2^0 (∘)
2	He	4	C_6	1:1:0 :: 6	196	1	8	205	18	18	0, 0	Seated ($E_2^\pm$)
3	Li	7	C_6	2:1:1 :: 6	362	1	12	375	18	18	0, 0	Seated (H_μ turn)
4	Be	9	C_6	2:2:0 :: 6	460	1	16	477	18	18	0, 0	Seated ($E_2^\pm$)
5	B	11	C_6	3:2:1 :: 6	558	1	20	579	18	18	0, 0	Seated ($E_2^\pm$)
6	C	12	C_6	3:3:0 :: 6	588	1	24	613	18	18	0, 0	Filled Anchor ⧫
7	N	14	C_8	1:0:1 :: 8	686	1	4	691	24	24	0, 0	Seated ($E_2^\pm$)
8	O	16	C_8	1:1:0 :: 8	784	1	8	793	21	24	−3, 0	Acceptor E_2^0 (∘)
9	F	19	C_8	2:1:1 :: 8	950	1	12	963	21	24	−3, 0	Acceptor E_2^0 (∘)
10	Ne	20	C_8	2:2:0 :: 8	980	1	16	997	24	24	0, 0	Seated (H_μ turn)
11	Na	23	C_8	3:2:1 :: 8	1146	1	20	1167	24	24	0, 0	Seated ($E_2^\pm$)
12	Mg	24	C_8	3:3:0 :: 8	1176	1	24	1201	24	24	0, 0	Seated ($E_2^\pm$)
13	Al	27	C_8	4:3:1 :: 8	1342	1	28	1371	24	24	0, 0	Seated ($E_2^\pm$)
14	Si	28	C_8	4:4:0 :: 8	1372	1	32	1405	24	24	0, 0	Filled Anchor ⧫
15	P	31	C_{10}	1:0:1 :: 10	1538	1	4	1543	30	30	0, 0	Seated ($E_2^\pm$)
16	S	32	C_{10}	1:1:0 :: 10	1568	1	8	1577	27	30	−3, 0	Acceptor E_2^0 (∘)
17	Cl	35	C_{10}	2:1:1 :: 10	1734	1	12	1747	27	30	−3, 0	Acceptor E_2^0 (∘)
18	Ar	40	C_{10}	2:2:0 :: 10	2036	1	16	2053	30	30	0, 0	Seated ($E_2^\pm$)
19	K	39	C_{10}	3:2:1 :: 10	1930	1	20	1951	30	30	0, 0	Seated (H_μ turn)
20	Ca	40	C_{10}	3:3:0 :: 10	1960	1	24	1985	30	30	0, 0	Seated ($E_2^\pm$)
21	Sc	45	C_{10}	4:3:1 :: 10	2262	1	28	2291	30	30	0, 0	Seated ($E_2^\pm$)
22	Ti	48	C_{10}	4:4:0 :: 10	2428	1	32	2461	30	30	0, 0	Seated ($E_2^\pm$)
23	V	51	C_{10}	5:4:1 :: 10	2594	1	36	2631	30	30	0, 0	Seated ($E_2^\pm$)
24	Cr	52	C_{10}	5:5:0 :: 10	2624	1	40	2665	30	30	0, 0	Filled Anchor ⧫
25	Mn	55	C_{12}	1:0:1 :: 12	2790	1	4	2795	36	36	0, 0	Seated ($E_2^\pm$)
26	Fe	56	C_{12}	1:1:0 :: 12	2820	1	8	2829	36	36	0, 0	Seated ($E_2^\pm$)
27	Co	59	C_{12}	2:1:1 :: 12	2986	1	12	2999	36	36	0, 0	Seated ($E_2^\pm$)
28	Ni	58	C_{12}	2:2:0 :: 12	2880	1	16	2897	36	36	0, 0	Seated ($E_2^\pm$)
29	Cu	63	C_{12}	3:2:1 :: 12	3182	1	20	3203	36	36	0, 0	Seated ($E_2^\pm$)
30	Zn	64	C_{12}	3:3:0 :: 12	3212	1	24	3237	36	36	0, 0	Seated (H_μ turn)
31	Ga	69	C_{12}	4:3:1 :: 12	3514	1	28	3543	36	36	0, 0	Seated ($E_2^\pm$)
32	Ge	74	C_{12}	4:4:0 :: 12	3816	1	32	3849	36	36	0, 0	Seated ($E_2^\pm$)
33	As	75	C_{12}	5:4:1 :: 12	3846	1	36	3883	36	36	0, 0	Seated ($E_2^\pm$)
34	Se	80	C_{12}	5:5:0 :: 12	4148	1	40	4189	33	36	−3, 0	Acceptor E_2^0 (∘)
35	Br	79	C_{12}	6:5:1 :: 12	4042	1	44	4087	33	36	−3, 0	Acceptor E_2^0 (∘)
36	Kr	84	C_{12}	6:6:0 :: 12	4344	1	48	4393	36	36	0, 0	Filled Anchor ⧫
37	Rb	85	C_{14}	1:0:1 :: 14	4374	1	4	4379	42	42	0, 0	Seated ($E_2^\pm$)
38	Sr	88	C_{14}	1:1:0 :: 14	4540	1	8	4549	42	42	0, 0	Seated ($E_2^\pm$)
39	Y	89	C_{14}	2:1:1 :: 14	4570	1	12	4583	42	42	0, 0	Seated ($E_2^\pm$)
40	Zr	90	C_{14}	2:2:0 :: 14	4600	1	16	4617	42	42	0, 0	Seated ($E_2^\pm$)
41	Nb	93	C_{14}	3:2:1 :: 14	4766	1	20	4787	42	42	0, 0	Seated ($E_2^\pm$)
42	Mo	98	C_{14}	3:3:0 :: 14	5068	1	24	5093	42	42	0, 0	Seated ($E_2^\pm$)
43	Tc	98	C_{14}	4:3:1 :: 14	5030	1	28	5059	45	42	3, 3	Exoshed ↑ C_{16}
44	Ru	102	C_{14}	4:4:0 :: 14	5264	1	32	5297	42	42	0, 0	Seated ($E_2^\pm$)
45	Rh	103	C_{14}	5:4:1 :: 14	5294	1	36	5331	42	42	0, 0	Seated ($E_2^\pm$)
46	Pd	106	C_{14}	5:5:0 :: 14	5460	1	40	5501	42	42	0, 0	Seated ($E_2^\pm$)
47	Ag	107	C_{14}	6:5:1 :: 14	5490	1	44	5535	42	42	0, 0	Seated ($E_2^\pm$)
48	Cd	114	C_{14}	6:6:0 :: 14	5928	1	48	5977	42	42	0, 0	Seated ($E_2^\pm$)
49	In	115	C_{14}	7:6:1 :: 14	5958	1	52	6011	42	42	0, 0	Seated ($E_2^\pm$)
50	Sn	120	C_{14}	7:7:0 :: 14	6260	1	56	6317	42	42	0, 0	Filled Anchor ⧫
51	Sb	121	C_{16}	1:0:1 :: 16	6290	1	4	6295	48	48	0, 0	Seated ($E_2^\pm$)
52	Te	130	C_{16}	1:1:0 :: 16	6864	1	8	6873	45	48	−3, 0	Acceptor E_2^0 (∘)
53	I	127	C_{16}	2:1:1 :: 16	6622	1	12	6635	45	48	−3, 0	Acceptor E_2^0 (∘)
54	Xe	132	C_{16}	2:2:0 :: 16	6924	1	16	6941	48	48	0, 0	Seated ($E_2^\pm$)
55	Cs	133	C_{16}	3:2:1 :: 16	6954	1	20	6975	48	48	0, 0	Seated ($E_2^\pm$)
56	Ba	138	C_{16}	3:3:0 :: 16	7256	1	24	7281	48	48	0, 0	Seated ($E_2^\pm$)
57	La	139	C_{16}	4:3:1 :: 16	7286	1	28	7315	48	48	0, 0	Seated ($E_2^\pm$)

Z	Sym	A	C_{2m}	$a{:}b{:}c :: L$	Π^{-}	H_{μ}	Π^{+}	RD_{fuse}	P^{D}	C_{close}	$S_{n+1}, X_{\mathrm{shed}}$	Route / Status
58	Ce	140	C_{16}	4:4:0 :: 16	7316	1	32	7349	48	48	0, 0	Seated (H_{μ} turn)
59	Pr	141	C_{16}	5:4:1 :: 16	7346	1	36	7383	48	48	0, 0	Seated ($E_2^{\pm}$)
60	Nd	142	C_{16}	5:5:0 :: 16	7376	1	40	7417	48	48	0, 0	Seated ($E_2^{\pm}$)
61	Pm	145	C_{16}	6:5:1 :: 16	7542	1	44	7587	51	48	3, 3	Exoshed $\uparrow \mathsf{C}_{18}$
62	Sm	152	C_{16}	6:6:0 :: 16	7980	1	48	8029	48	48	0, 0	Seated ($E_2^{\pm}$)
63	Eu	153	C_{16}	7:6:1 :: 16	8010	1	52	8063	48	48	0, 0	Seated ($E_2^{\pm}$)
64	Gd	158	C_{16}	7:7:0 :: 16	8312	1	56	8369	48	48	0, 0	Seated ($E_2^{\pm}$)
65	Tb	159	C_{16}	8:7:1 :: 16	8342	1	60	8403	48	48	0, 0	Seated ($E_2^{\pm}$)
66	Dy	164	C_{16}	8:8:0 :: 16	8644	1	64	8709	48	48	0, 0	Filled Anchor ♦
67	Ho	165	C_{18}	1:0:1 :: 18	8674	1	4	8679	54	54	0, 0	Seated ($E_2^{\pm}$)
68	Er	166	C_{18}	1:1:0 :: 18	8704	1	8	8713	54	54	0, 0	Seated ($E_2^{\pm}$)
69	Tm	169	C_{18}	2:1:1 :: 18	8870	1	12	8883	54	54	0, 0	Seated ($E_2^{\pm}$)
70	Yb	174	C_{18}	2:2:0 :: 18	9172	1	16	9189	54	54	0, 0	Seated ($E_2^{\pm}$)
71	Lu	175	C_{18}	3:2:1 :: 18	9202	1	20	9223	54	54	0, 0	Seated ($E_2^{\pm}$)
72	Hf	180	C_{18}	3:3:0 :: 18	9504	1	24	9529	54	54	0, 0	Seated ($E_2^{\pm}$)
73	Ta	181	C_{18}	4:3:1 :: 18	9534	1	28	9563	54	54	0, 0	Seated ($E_2^{\pm}$)
74	W	184	C_{18}	4:4:0 :: 18	9700	1	32	9733	54	54	0, 0	Seated ($E_2^{\pm}$)
75	Re	187	C_{18}	5:4:1 :: 18	9866	1	36	9903	54	54	0, 0	Seated (H_{μ} turn)
76	Os	192	C_{18}	5:5:0 :: 18	10168	1	40	10209	54	54	0, 0	Seated ($E_2^{\pm}$)
77	Ir	193	C_{18}	6:5:1 :: 18	10198	1	44	10243	54	54	0, 0	Seated ($E_2^{\pm}$)
78	Pt	195	C_{18}	6:6:0 :: 18	10296	1	48	10345	54	54	0, 0	Seated ($E_2^{\pm}$)
79	Au	197	C_{18}	7:6:1 :: 18	10394	1	52	10447	54	54	0, 0	Seated ($E_2^{\pm}$)
80	Hg	202	C_{18}	7:7:0 :: 18	10696	1	56	10753	54	54	0, 0	Seated ($E_2^{\pm}$)
81	Tl	205	C_{18}	8:7:1 :: 18	10862	1	60	10923	54	54	0, 0	Seated ($E_2^{\pm}$)
82	Pb	208	C_{18}	8:8:0 :: 18	11028	1	64	11093	54	54	0, 0	Seated ($E_2^{\pm}$)
83	Bi	209	C_{18}	9:8:1 :: 18	11058	1	68	11127	54	54	0, 0	Seated ($E_2^{\pm}$)
84	Po	209	C_{18}	9:9:0 :: 18	11020	1	72	11093	57	54	3, 3	Anchor ♦ + $\uparrow \mathsf{C}_{20}$
85	At	210	C_{20}	1:0:1 :: 20	11050	1	4	11055	63	60	3, 3	Exoshed $\uparrow \mathsf{C}_{22}$
86	Rn	222	C_{20}	1:1:0 :: 20	11828	1	8	11837	63	60	3, 3	Exoshed $\uparrow \mathsf{C}_{22}$
87	Fr	223	C_{20}	2:1:1 :: 20	11858	1	12	11871	63	60	3, 3	Exoshed $\uparrow \mathsf{C}_{22}$
88	Ra	226	C_{20}	2:2:0 :: 20	12024	1	16	12041	63	60	3, 3	Exoshed $\uparrow \mathsf{C}_{22}$
89	Ac	227	C_{20}	3:2:1 :: 20	12054	1	20	12075	63	60	3, 3	Exoshed $\uparrow \mathsf{C}_{22}$
90	Th	232	C_{20}	3:3:0 :: 20	12356	1	24	12381	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
91	Pa	231	C_{20}	4:3:1 :: 20	12250	1	28	12279	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
92	U	238	C_{20}	4:4:0 :: 20	12688	1	32	12721	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
93	Np	237	C_{20}	5:4:1 :: 20	12582	1	36	12619	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
94	Pu	244	C_{20}	5:5:0 :: 20	13020	1	40	13061	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
95	Am	243	C_{20}	6:5:1 :: 20	12914	1	44	12959	66	60	6, 6	Exoshed $\uparrow \mathsf{C}_{22}$
96	Cm	247	C_{20}	6:6:0 :: 20	13148	1	48	13197	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
97	Bk	247	C_{20}	7:6:1 :: 20	13110	1	52	13163	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
98	Cf	251	C_{20}	7:7:0 :: 20	13344	1	56	13401	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
99	Es	252	C_{20}	8:7:1 :: 20	13374	1	60	13435	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
100	Fm	257	C_{20}	8:8:0 :: 20	13676	1	64	13741	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
101	Md	258	C_{20}	9:8:1 :: 20	13706	1	68	13775	69	60	9, 9	Exoshed $\uparrow \mathsf{C}_{22}$
102	No	259	C_{20}	9:9:0 :: 20	13736	1	72	13809	72	60	12, 12	Exoshed $\uparrow \mathsf{C}_{22}$
103	Lr	266	C_{20}	10:9:1 :: 20	14174	1	76	14251	72	60	12, 12	Exoshed $\uparrow \mathsf{C}_{22}$
104	Rf	267	C_{20}	10:10:0 :: 20	14204	1	80	14285	72	60	12, 12	Anchor ♦ + $\uparrow \mathsf{C}_{22}$
105	Db	268	C_{22}	1:0:1 :: 22	14234	1	4	14239	78	66	12, 12	Exoshed $\uparrow \mathsf{C}_{24}$
106	Sg	269	C_{22}	1:1:0 :: 22	14264	1	8	14273	78	66	12, 12	Exoshed $\uparrow \mathsf{C}_{24}$
107	Bh	270	C_{22}	2:1:1 :: 22	14294	1	12	14307	78	66	12, 12	Exoshed $\uparrow \mathsf{C}_{24}$
108	Hs	277	C_{22}	2:2:0 :: 22	14732	1	16	14749	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
109	Mt	278	C_{22}	3:2:1 :: 22	14762	1	20	14783	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
110	Ds	281	C_{22}	3:3:0 :: 22	14928	1	24	14953	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
111	Rg	282	C_{22}	4:3:1 :: 22	14958	1	28	14987	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
112	Cn	285	C_{22}	4:4:0 :: 22	15124	1	32	15157	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
113	Nh	286	C_{22}	5:4:1 :: 22	15154	1	36	15191	81	66	15, 15	Exoshed $\uparrow \mathsf{C}_{24}$
114	Fl	289	C_{22}	5:5:0 :: 22	15320	1	40	15361	84	66	18, 18	Exoshed $\uparrow \mathsf{C}_{24}$
115	Mc	290	C_{22}	6:5:1 :: 22	15350	1	44	15395	84	66	18, 18	Exoshed $\uparrow \mathsf{C}_{24}$
116	Lv	293	C_{22}	6:6:0 :: 22	15516	1	48	15565	84	66	18, 18	Exoshed $\uparrow \mathsf{C}_{24}$
117	Ts	294	C_{22}	7:6:1 :: 22	15546	1	52	15599	84	66	18, 18	Exoshed $\uparrow \mathsf{C}_{24}$
118	Og	294	C_{22}	7:7:0 :: 22	15508	1	56	15565	84	66	18, 18	Exoshed $\uparrow \mathsf{C}_{24}$
119	–	297	C_{22}	8:7:1 :: 22	15674	1	60	15735	69	66	3, 3	Cutoff (C_{24} open)
120	–	298	C_{22}	8:8:0 :: 22	15704	1	64	15769	72	66	6, 6	Cutoff (C_{24} open)
121	–	301	C_{22}	9:8:1 :: 22	15870	1	68	15939	75	66	9, 9	Cutoff (C_{24} open)
122	–	302	C_{22}	9:9:0 :: 22	15900	1	72	15973	78	66	12, 12	Cutoff (C_{24} open)
123	–	305	C_{22}	10:9:1 :: 22	16066	1	76	16143	81	66	15, 15	Cutoff (C_{24} open)

Z	Sym	A	C_{2m}	$a{:}b{:}c :: L$	Π^-	H_μ	Π^+	RD_{fuse}	P^D	C_{close}	S_{n+1}, X_{shed}	Route / Status
124	–	306	C_{22}	10:10:0 :: 22	16096	1	80	16177	84	66	18, 18	Cutoff (C_{24} open)
125	–	309	C_{22}	11:10:1 :: 22	16262	1	84	16347	87	66	21, 21	Cutoff (C_{24} open)
126	–	310	C_{22}	11:11:0 :: 22	16292	1	88	16381	90	66	24, 24	Cutoff (C_{24} open)

5 Mandatory Audit Hooks, D.E.C. Verification, and Falsification

5.1 Execution of the six mandatory Main §5.17 audit hooks

Setup. Every row of the Matter I ledger (Table 3) is subjected to the six structural audit hooks imported from Main §5.17 and executed as automated exact-arithmetic assertions in `scripts/verify_matter1_ledger.py`.

Derivation. For each coord-0 slot $Z \in \{1, \dots, 126\}$:

1. **Spine audit (Main §5.17.1):** Checks that every retained occurrence at every finite depth has explicit branch/hinge/branch incidence $(\Pi^-_{\widehat{F}}, H_{\mu,\widehat{F},\widehat{G}}, \Pi^+_{\widehat{G}})$ with $\Pi^-_{\widehat{F}} > 0$, $H_{\mu,\widehat{F},\widehat{G}} \geq 1$, $\Pi^+_{\widehat{G}} > 0$, and $RD_{\text{fuse}} = \Pi^-_{\widehat{F}} + H_{\mu,\widehat{F},\widehat{G}} + \Pi^+_{\widehat{G}}$.
2. **Seat/enclosure routing audit (Main §5.17.4):** Checks that field-count scaling $\mathcal{O}_{\partial F}(B) = n_F(B) \log n_F(B)$ (Main (90)) and seat-compatible shell/enclosure routing ($\Gamma_{m-1} < Z \leq \Gamma_m$, $a{:}b{:}c :: 2m$, $X_{\mathcal{C}}(B)$) remain distinct calculations on the same declared boundary/window B_Z.
3. **Sheddic path audit (Main §5.17.5):** Checks that whenever $X_{\text{shedding}}(B_Z) > 0$, a sheddic path is explicitly declared through $\{E_2, \mathsf{C}_6, \mathsf{C}_8, \mathsf{C}_{10}, \dots, \mathsf{C}_{2n}\}$ and the seat-compatible retention condition $\text{TTL}_{\mathcal{C}}(\partial F, \omega) < |\omega|$ is recorded; whenever $X_{\text{shedding}}(B_Z) = 0$, $\text{TTL}_{\mathcal{C}}(\partial F, \omega) \geq |\omega|$ holds.
4. **Cycle audit (Main §5.17.7):** Checks that the certified cycle-valued field compression $C_f[a{:}b{:}c :: 2m]_{A \leftrightarrow \Omega}$ (Main (40)) is constructed and bound before any scalar duration accessor $RD(C_f[p{:}q])$ is evaluated.
5. **Fusion closure audit (Main §5.17.8):** Checks that admitted curling-curl support forms $\widehat{F}_n, \widehat{G}_n$, hinge μ, closure scope B_{n+1}, fusion duration RD_{fuse}, and signed rung residual $S_{n+1} = P^D_{n+1} - C_{\text{close},n+1}$ are declared before forming $X_{\text{shedding},n+1} = \max(0, S_{n+1})$ or any exoshedding lift K^{r+1}_{prop}.
6. **Naming audit (Main §5.17.10):** Checks that every label identifies its declared distinction, relation, motif, or dynamic, while measured-sector carrier symbols (H…Og for $Z = 1 \dots 118$, and $--$ for unretained cutoff slots $Z = 119 \dots 126$) remain restricted to the application layer.

Specialization. Table 4 records the verified pass counts across all 126 coord-0 rows (118 admitted elements plus 8 unretained C_{22} cutoff slots).

Table 4: **Automated verification summary of Main §5.17 audit hooks (`scripts/verify_matter1_ledger.py`).**

Audit hook	Reference	Verified invariant and pass count
Spine audit	Main §5.17.1	**PASS** (126/126 rows verify $\Pi^- + H_\mu + \Pi^+ = RD_{\text{fuse}}$ with $H_\mu = 1$)
Seat/enclosure routing audit	Main §5.17.4	**PASS** (9/9 $\Gamma_m = m(m+1)-6$ boundaries and $m{:}m{:}0 :: 2m$ anchors verified distinct from $n_F \log n_F$)
Sheddic path audit	Main §5.17.5	**PASS** (37/37 admitted shedding rows and 8/8 cutoff rows declare SheddicPath $\in \{\mathsf{C}_{16} \dots \mathsf{C}_{24}\}$ and $\text{TTL}_{\mathcal{C}} < \lvert\omega\rvert$)
Cycle audit	Main §5.17.7	**PASS** (126/126 rows bind $C_f[a{:}b{:}c :: 2m]_{A\leftrightarrow\Omega}$ prior to RD_{fuse})
Fusion closure audit	Main §5.17.8	**PASS** (126/126 rows verify $S_{n+1} = P^D_{n+1} - C_{\text{close},n+1}$ and $X_{\text{shedding},n+1} = \max(0, S_{n+1})$)
Naming audit	Main §5.17.10	**PASS** (118/118 element symbols quarantined as carrier labels; 8/8 cutoff slots marked $--$)

Calibration. In the D.E.C. (Declared Edge Computation) ledger representation (Manual §4.1–§4.9), each elemental rung corresponds to a 9-tuple row

$$\big(B_Z,\ v,\ e,\ v_e,\ \sigma(e),\ \text{adm}(e;B_Z),\ w(e;B_Z),\ P(e;B_Z),\ \text{route}(e;B_Z)\big)$$

satisfying $\sum_e P(e;B_Z) = 1$ on admissible support, $\text{adm}(e;B_Z) = 0 \implies P(e;B_Z) = 0$, and antisymmetric SADAR pairing $\mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e)$ (Main (95)).

Falsification. Any ledger row in which:

1. $RD_{\text{fuse}} \neq \Pi^-_{\widehat{F}} + H_{\mu,\widehat{F},\widehat{G}} + \Pi^+_{\widehat{G}}$,
2. Z falls outside $(\Gamma_{m-1}, \Gamma_m]$ for its declared C_{2m},
3. a filled shared enclosure at $Z = \Gamma_m$ fails $a = b = m, c = 0$,
4. a deficit acceptor ($S_{n+1} < 0$) is assigned positive X_{shedding}, or
5. a shedding row ($X_{\text{shedding}} > 0$) lacks a declared even enclosure route or violates $\text{TTL}_{\mathcal{C}} < \lvert\omega\rvert$

contradicts the rung construction and fails automated verification in `verify_matter1_ledger.py`.

Provenance. Main §5.17.1–§5.17.11, Main (40), Main (90), Main (95), Main (107)–(121); Manual §4.1–§4.9.

Literature note. D.E.C. ledger execution and audit invariants: Poyau [4] (Manual §4).

6 Prediction-Test Fixture Registry

Scope. This registry records completion fields for Matter I ($Z = 1 \dots 118$ and C_{22} cutoff $119 \dots 126$) fixtures and comparisons. Rows carry their data-support class, active fields, required AΩ calculation, and audit artifact.

Table 5: **Matter I prediction-test fixture registry.**

Record	Class	Active fields	Required AΩ calculation	Artifact
Curling-curl primitives (1:2, 3:1, 3:2, 3:3, 3:4)	D0	integer $p, q, H_\mu = 1$	$\Pi(p{:}q) = q^p + q$, $RD^{(1)}_{A\leftrightarrow\Omega} = 2\Pi + 1$	Table 1

Record	Class	Active fields	Required AΩ calculation	Artifact
Coord-0 enclosure groups ($m = 3 \dots 11$)	D0	integer $m, 2m, \Gamma_{m-1}, \Gamma_m$	$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6$, $m{:}m{:}0 :: 2m$	Table 2, `data/m1/`
Filled shared anchors ($Z = \Gamma_m \in \{6, \dots, 104\}$)	D0	integer $Z = \Gamma_m$, $a = b = m, c = 0$	$0 \vdash\!:: Z : A : 0 : 2m\langle m{:}m{:}0 :: 2m\rangle$, RD_{fuse}	Table 3, Fig. 1
Seated retained closures (66 non-anchor rungs)	D0	integer $Z, A, s, E_2^{\pm}$	$S_{n+1} = P^D_{n+1} - C_{\text{close},n+1} = 0$, $X_{\text{shedding}} = 0$, $\text{TTL}_{\mathcal{C}} \geq \lvert\omega\rvert$	Table 3
Open-seat acceptors (9 rungs: H, O..I)	D0	integer Z, A, s, E_2^0	$S_{n+1} = -3 < 0$ (deficit/open check), $X_{\text{shedding}} = 0$	Table 3
Mid-shell hinge exoshedding ($Z \in \{43, 61\}$)	D0	integer Z, A, s, odd-window Yro_{2m+1}	$S_{n+1} = +3 > 0$, $X_{\text{shedding}} = 3$, $\text{Lift}_{B_r \to B_{r+1}}$	Table 3
Post-bismuth shedding ladder ($Z = 84 \dots 118$)	D0	integer Z, A, s, Yro_{2m+1}	$S_{n+1} > 0$, $X_{\text{shedding}} > 0$, $\text{TTL}_{\mathcal{C}} < \lvert\omega\rvert$, $\text{SheddicPath} = \mathsf{C}_{2m+2}$	Table 3
Unretained C_{22} cutoff ($Z = 119 \dots 126$)	D0	integer $Z, s \in \{15 \dots 22\}$	$\text{TTL}_{\mathcal{C}} = 0 < \lvert\omega\rvert$, $X_{\text{shedding}} = X_{\text{open}} > 0$	Table 3
Four-step closure load ($L = 6 \dots 22$)	D0	integer $L = 4c_L + r_L$	$c_L = \lfloor L/4 \rfloor$, $r_L = L \bmod 4 \in \{0, 1, 2, 3\}$	App. A
Shell-seat parity and hemisphere δ_3 fixture	D0	integer comparator bins $O_k, \widehat{O}_k$	$\Delta_k^{\mathbb{Z}} = \widehat{O}_k - O_k$, $\delta_{3,k} = (s_k^{(3)}, m_k)$	App. C
Elemental mass and spectral target manifest	O0	quarantined target rows $\mathcal{T}_Z$	$\mathsf{Q}_{\text{ext}}(\nu_{Z,A\leftrightarrow\Omega}; \mathcal{T}_Z)$ after internal candidate freeze	App. B

Completion rule. Exact $D0$ rows record integer support, RD_{fuse}, P^D, C_{close}, S_{n+1}, X_{shedding}, and audit status. Quarantined $O0$ comparison rows attach external targets only after internal coord-0 declarations freeze.

A Simulation and Prediction-Test Fixtures

A.1 Scope and required outputs

This appendix preserves exact simulation fixtures and four-step closure reductions downstream from Main App. H and Main App. I. Required exact outputs for each elemental rung are

$$\Pi(p{:}q), \quad RD^{(1)}_{A\leftrightarrow\Omega}, \quad \Gamma_m, \quad RD_{\text{fuse}}, \quad P^D_{n+1}, \quad C_{\text{close},n+1}, \quad S_{n+1}, \quad X_{\text{shedding},n+1}.$$

A.2 Four-step arithmetic fixtures across $\mathsf{C}_6 \dots \mathsf{C}_{22}$

Specializing Main App. I.2 ($L = 4c_L + r_L$, $r_L \in \{0, 1, 2, 3\}$) to the nine outer even support enclosures $L = 2m \in \{6, 8, 10, 12, 14, 16, 18, 20, 22\}$ gives the exact integer decompositions:

$$\begin{aligned} 6 &= 4(1) + 2, & 8 &= 4(2) + 0, & 10 &= 4(2) + 2, \\ 12 &= 4(3) + 0, & 14 &= 4(3) + 2, & 16 &= 4(4) + 0, \\ 18 &= 4(4) + 2, & 20 &= 4(5) + 0, & 22 &= 4(5) + 2. \end{aligned}$$

Every even outer enclosure has $r_L \in \{0, 2\}$; odd-window Yro_{2m+1} routes between C_{2m} and C_{2m+2} have $r_{2m+1} \in \{1, 3\}$.

A.3 Exact rational occupancy invariants

Across $m = 3 \dots 11$, total coord-0 capacity is $\Gamma_{11} = 126$. The admitted elemental ladder $Z = 1 \dots 118$ and the terminal ring C_{22} (22 seats, 14 admitted, 8 cutoff) satisfy the exact rational identities verified

in `manual2/scripts/verify_matter1_ledger.py`:

$$\frac{Z_{\max}}{\Gamma_{11}} = \frac{118}{126} = \frac{59}{63}, \qquad \frac{N_{\mathrm{adm}}(\mathsf{C}_{22})}{22} = \frac{14}{22} = \frac{7}{11}, \qquad \frac{N_{\mathrm{cutoff}}(\mathsf{C}_{22})}{22} = \frac{8}{22} = \frac{4}{11}.$$

A.4 Falsification

A four-step decomposition with non-integer c_L or $r_L \notin \{0,1,2,3\}$, or an even enclosure with odd remainder $r_L \in \{1,3\}$, falsifies the fixture on that scope (Main App. I.4).

Provenance. Main §3.5, Main §5.7.13, Main §5.9.1, Main App. H.5, Main App. I.2.

B Quarantined External Comparison

B.1 Scope and presentation-numeric policy

This appendix compares AΩ outputs against external measured-sector coordinates without importing external units, constants, or decimals into the AΩ calculation (Manual App. B). An active Matter I calculation produces an exact integer or rational candidate $\nu_{Z,A\leftrightarrow\Omega}$ first; only after that candidate is frozen may an external comparison record evaluate

$$\mathsf{Q}_{\mathrm{ext}}\big(\nu_{Z,A\leftrightarrow\Omega};\ \mathcal{T}_Z\big), \tag{22}$$

where $\mathcal{T}_Z$ is a declared measured-sector target record (such as an AME atomic mass record [5] or NIST ASD spectral/ionization record [1]). Every released decimal presentation value carries the 6-tuple precision record (Manual §1.7.1)

$$\big(x_{\mathrm{disp}},\ u_x,\ \mathrm{unit}(x),\ \mathrm{src}(x),\ \mathrm{round}(x),\ \mathrm{scope}(x)\big). \tag{23}$$

B.2 Quarantine audit hooks (F1–F5)

- **F1.** External-comparison audit records the active $\nu_{Z,A\leftrightarrow\Omega}$ candidate before the external comparison.
- **F2.** Speed/transition-ratio audit records exact integer or rational $A\leftrightarrow\Omega$ ratios before wavelength or energy conversion.
- **F3.** Direction-cycle audit records branch/hinge orientation $\chi_\mu \in \{-1,+1\}$ relative to one complete closure cycle 1_C.
- **F4.** Charge/seat-comparison audit records integer coord-0 index Z and seat status (E_2^0 vs. $E_2^\pm$) before external charge symbols.
- **F5.** Quarantine audit records external constants $(c, h_{\mathrm{ext}}, e, \eta_{\mathrm{ext}})$ and measured units $(\mathrm{MeV}, \mathrm{eV}, \mathrm{nm})$ only after the active AΩ calculation.

Provenance. Main (79), Main (98); Manual §1.7, §1.8, App. B.

Literature note. Atomic mass and spectral target databases: Wang et al. [5]; NIST ASD [1].

C Coord-0 Shell-Seat Phase-Cycle δ_3 Fixture Specification

C.1 Active integer data form and hemisphere/parity comparators

This appendix records the exact integer δ_3 comparator specification (Manual §1.7.1, App. C) for the nine concentric coord-0 support enclosures C_{2m} ($m = 3 \dots 11$). On each ring C_{2m} of $2m$ seats:

1. The **parity count** partitions seats $s \in \{1, \dots, 2m\}$ into odd open-seat steps ($c = 1$: $O_{\text{odd},m} = m$) and even duad-closed steps ($c = 0$: $O_{\text{even},m} = m$). Against the balanced integer comparator $\widehat{O}_m = m$, the exact residual is

$$\Delta^{\mathbb{Z}}_{\text{parity},m} = \widehat{O}_m - O_{\text{even},m} = 0, \qquad \delta_{3,\text{parity},m} = (0, 0).$$

2. The **hemisphere count** partitions the $2m$ seats into outbound Π_α seats ($s \in \{1, \dots, m-1\}$, count $m-1$), the 6 o'clock H_μ turnaround seat ($s = m$, count 1), return Π_Ω seats ($s \in \{m+1, \dots, 2m-1\}$, count $m-1$), and the 12 o'clock filled enclosure spine seat ($s = 2m$, count 1). Against the symmetric branch comparator $\widehat{\Pi}_m = m - 1$,

$$\Delta^{\mathbb{Z}}_{\text{branch},m} = |\Pi_{\alpha,m}| - |\Pi_{\Omega,m}| = (m-1) - (m-1) = 0, \qquad \delta_{3,\text{branch},m} = (0, 0).$$

3. On the terminal ring C_{22} ($m = 11$, 22 seats), comparing the admitted element count $T^{\times}_{\mathsf{C}_{22}} = 14$ ($Z = 105 \dots 118$) against the full ring seat capacity $O_{\mathsf{C}_{22}} = 22$ gives the exact cutoff residual

$$\Delta^{\mathbb{Z}}_{\mathsf{C}_{22}} = T^{\times}_{\mathsf{C}_{22}} - O_{\mathsf{C}_{22}} = 14 - 22 = -8, \qquad \delta_{3,\mathsf{C}_{22}} = (-1, 8),$$

recording the 8 unretained open cutoff slots $Z = 119 \dots 126$.

C.2 Final lock

All δ_3 tallies in this volume use exact integer counts $\Delta^{\mathbb{Z}}_k = s^{(3)}_k m_k$ with $s^{(3)}_k \in \{-1, 0, +1\}$ and $m_k = |\Delta^{\mathbb{Z}}_k| \in \mathbb{Z}_{\geq 0}$.

Provenance. Main §5.7.10–§5.7.13; Manual §1.7.1 (RG10), App. C.

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